GATE-AIII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 09



By- Rahul sir

Recap of previous lecture



Topic Rolles theorem

Topic Lagrange mean value theorem

Topic Problems based on mean value theorem

Topic Basic integration

Topics to be covered



- **Topic** Integration via parts
- **Topic** Definite integrals
- **Topic** Improper integrals
- **Topic** Area bounded regions

LECTURE-10
Single variable
Calculus.
Maclaurins/Taylor

• Rahul Sir

Integration Via Parts:

$$\int \text{product of Two functions} = \text{Integration Via Parts}$$

$$\int \underset{\text{I}}{f(x)}\,\underset{\text{II}}{g(x)}\,dx = f(x)\cdot\int g(x)\,dx - \int\left[\frac{d}{dx}f(x)\int g(x)\,dx\right]dx$$

$$\underset{\text{II}}{f(x)}\quad\underset{\text{I}}{g(x)}$$

ILATE Rule

- I → Inverse circular function
- L → Logarithmic functions
- A → Algebraic Function
- T → Trigonometric function
- E → Exponetial functions

$$\int \underset{\text{II}}{e^x}\,\underset{\text{I}}{\log x}\,dx \qquad \int \underset{\text{II}}{e^x}\,\underset{\text{I}}{\sin x}$$

$$\int x\tan x\,dx = \int (\text{I}\times\text{II})\,dx$$

$$I = \int \underset{II}{e^x} \underset{I}{\sin x} = \int (I \times II)\, dx$$

$$= I \int II\, dx - \int \left[\frac{d}{dx}(I) \int II\, dx \right] dx$$

$$= \sin x \int e^x dx - \int \left[\frac{d}{dx}(\sin x) \int e^x dx \right] dx$$

$$= e^x \sin x - \int \underline{\cos x}\, \underline{e^x}\, dx$$

$$= e^x \sin x - \left[\cos x \int e^x dx - \int \sin x\, e^x dx \right]$$

$$I = e^x \sin x - \cos x\, e^x - I$$

$$2I = e^x \sin x - \cos x\, e^x$$

$$\boxed{I = \frac{e^x}{2}\left[\sin x - \cos x\right]}$$

x^2, x^3, x^4] algebraic functions

only 2 step Using Tricks

$$\int \log x \, dx = \int \boxed{t e^t \, dt}$$

Vsmg Integration via Parts

$$= t\int e^t \, dt - \int \left[\frac{d}{dt}(t)\int e^t \, dt\right] dt$$

$$= t e^t - e^t$$

$$= e^t(t-1) = e^{\log_e x}(\log x - 1) = \underline{x(\log x - 1)}$$

Strategy $\Rightarrow$ Vsmg

$\log x = t$

$x = e^t$

$\boxed{dx = e^t \, dt}$

Vsmg Trick : Integration Via Parts :

$$\int (\underbrace{\text{algebraic function}}_{\text{must be}}) \times (\text{Another function}) \, dx$$

1) $I = \int x^2 e^x \, dx$

Desi | Integration

x^2 (+) e^x

$2x$ (−) e^x

2 (+) e^x

0 e^x

STOP

(Desivative Becomes ZERO)

$$I = x^2e^x - 2xe^x + 2e^x$$ Ans

1) x^2 = Algebraic function

algebraic function — desivative becomes ZERO

DERIVATIVE

2) another function — Integration

3) $\frac{d}{dx}(x^2) = 2x$

$x^n = nx^{n-1}$

$= 2x^{2-1}$

$= 2x^1$

$= 2x$

2) $I = \int (\log x)^2 x^2 \, dx$

$\log x = t$

$x = e^t \quad dx = e^t dt$

$= \int t^2 e^{2t} \cdot e^t dt$

$= \int t^2 e^{3t} dt$

t^2	e^{3t}	
$2t$	$e^{3t}/3$	
2	$e^{3t}/9$	
0	$\frac{e^{3t}}{27}$	

$= t^2\frac{e^{3t}}{3} - 2t\frac{e^{3t}}{3\times3} + 2\frac{e^{3t}}{9\times3}$

3) $I = \int x^2 \cos x \, dx$

Derivative		Integ
x^2	+	$\cos x$
$2x$	−	$+\sin x$
2	+	$-\cos x$
0		$-\sin x$

$$\boxed{I = x^2 \sin x + 2x\cos x - 2\sin x}$$

4) $I = \int x^3 e^{3x} \, dx$

$$I = x^3\frac{e^{3x}}{3} - 3x^2\frac{e^{3x}}{9} + 6x\frac{e^{3x}}{27} - \frac{6e^{3x}}{81}$$

5) $I = \int (x^2+5x+6) \sin x \, dx$

D I

(x^2+5x+6)	$\sin x$
$2x+5$	$-(\cos x)$
2	$-\sin x$
0	$\cos x$

5-Talk
= UNNECESSARY

CASE02

$$I = \int e^x \sin x\,dx \text{ or } \int e^x \cos x\,dx$$

only for

D | I

e^x | $\sin x$ (+)

e^x | $-\cos x$ (=)

e^x | $-\sin x$

repEAT - stop

Multiply with Integral Sign $\int$

$$I = -e^x \cos x + e^x \sin x = \int e^x \sin x\,dx$$

$$2I = e^x \sin x - e^x \cos x$$

$$\boxed{I = \frac{e^x \sin x - e^x \cos x}{2}}$$

Trigonometric Function
= sin / cos
= Integration

other function (e^x)
derivative

$$I = \int e^x \cos x\,dx$$

$$I = \int e^{ax} \sin bx\,dx$$

$$I = \int e^x \cos x \, dx$$

D	I
e^x	$\cos x$
e^x	$+\sin x$
e^x	$-\cos x$

$$I = \frac{e^x \sin x + e^x \cos x}{2}$$

$I = \int e^{ax} \sin bx \, dx$

$I = \int e^{ax} \cos bx \, dx$

Definite Integral: (A) $\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$

$\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

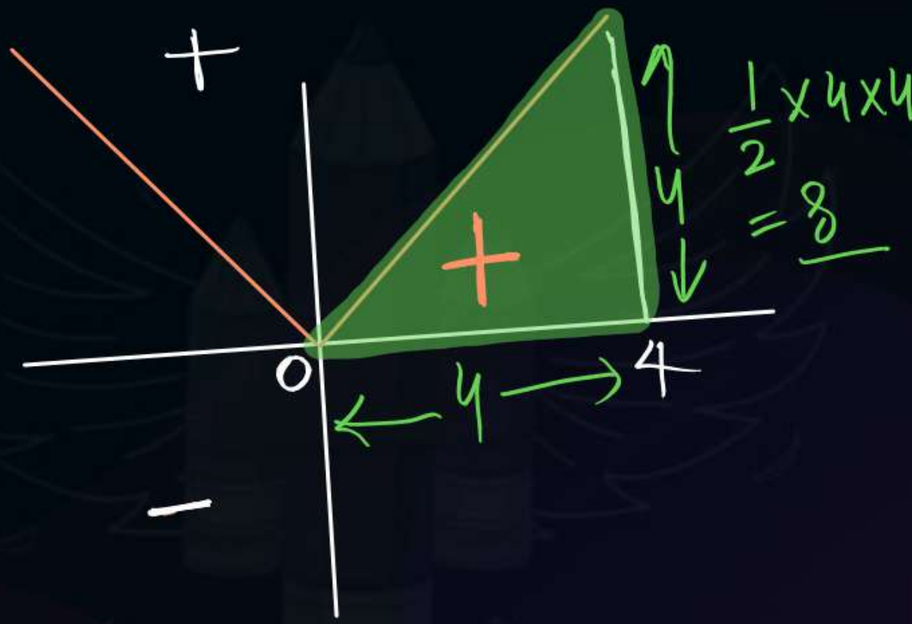
$\int_0^{\frac{\pi}{2}} \sin x\,dx = [-\cos x]_0^{\pi/2} = -\cos\frac{\pi}{2} - (-\cos 0) = 1$

$\int_0^4 |x|\,dx$ $\quad |x| \longrightarrow 0 \text{ to } 4$, $|x| = +$

$|x|$ = Compound Function

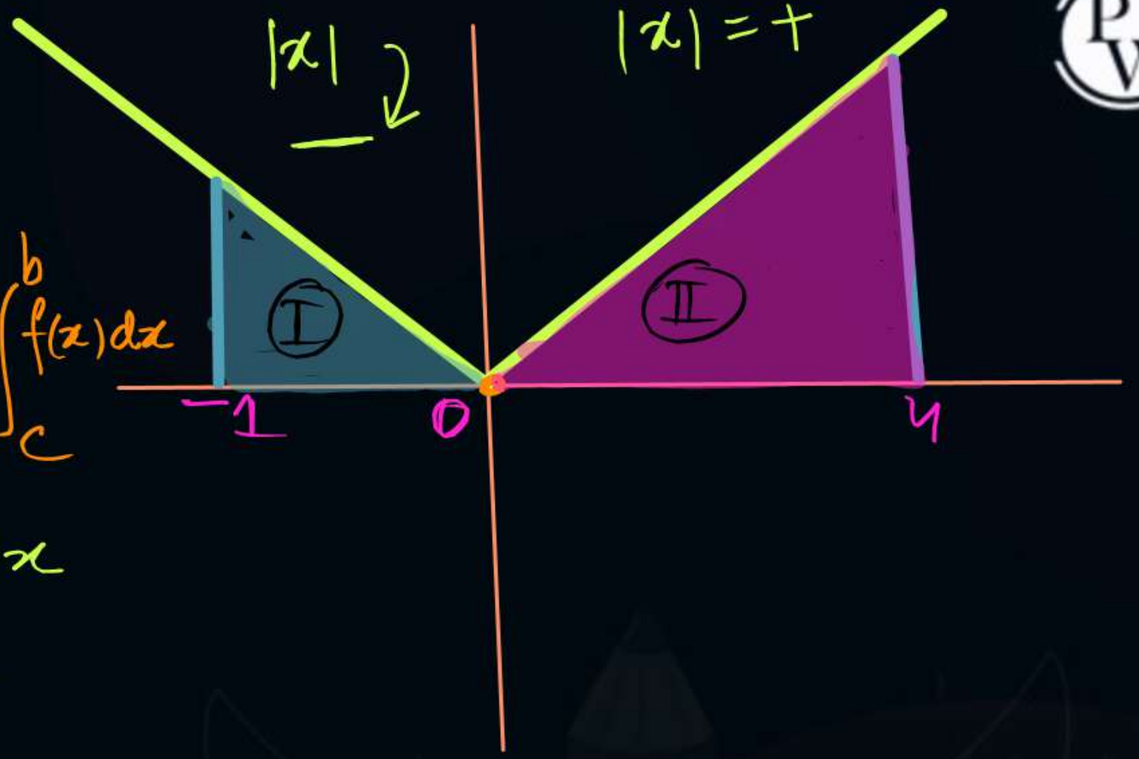
$= \int_0^4 x\,dx = \left[\frac{x^2}{2}\right]_0^4$

$= \frac{4^2}{2} = \underline{8}$



$$\int_{-1}^{4} |x|\,dx$$

Property No 1 : $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

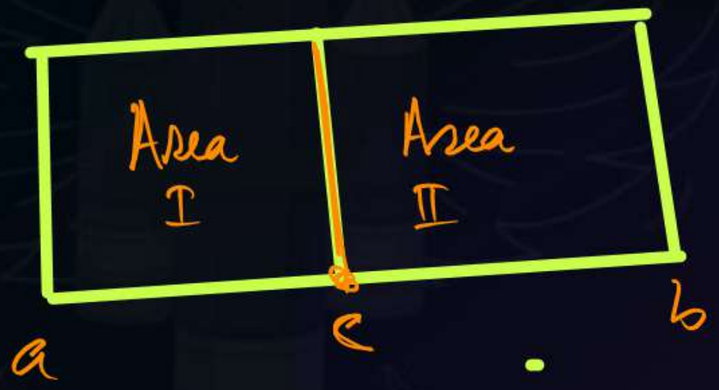


$$\int_{-1}^{4} |x|\,dx = \int_{-1}^{0} -x\,dx + \int_{0}^{4} x\,dx$$

$$= \left[-\frac{x^2}{2}\right]_{-1}^{0} + \left[\frac{x^2}{2}\right]_{0}^{4}$$

$$= +\frac{1}{2} + 8$$

$$= \frac{17}{2} \quad \text{Ans}$$



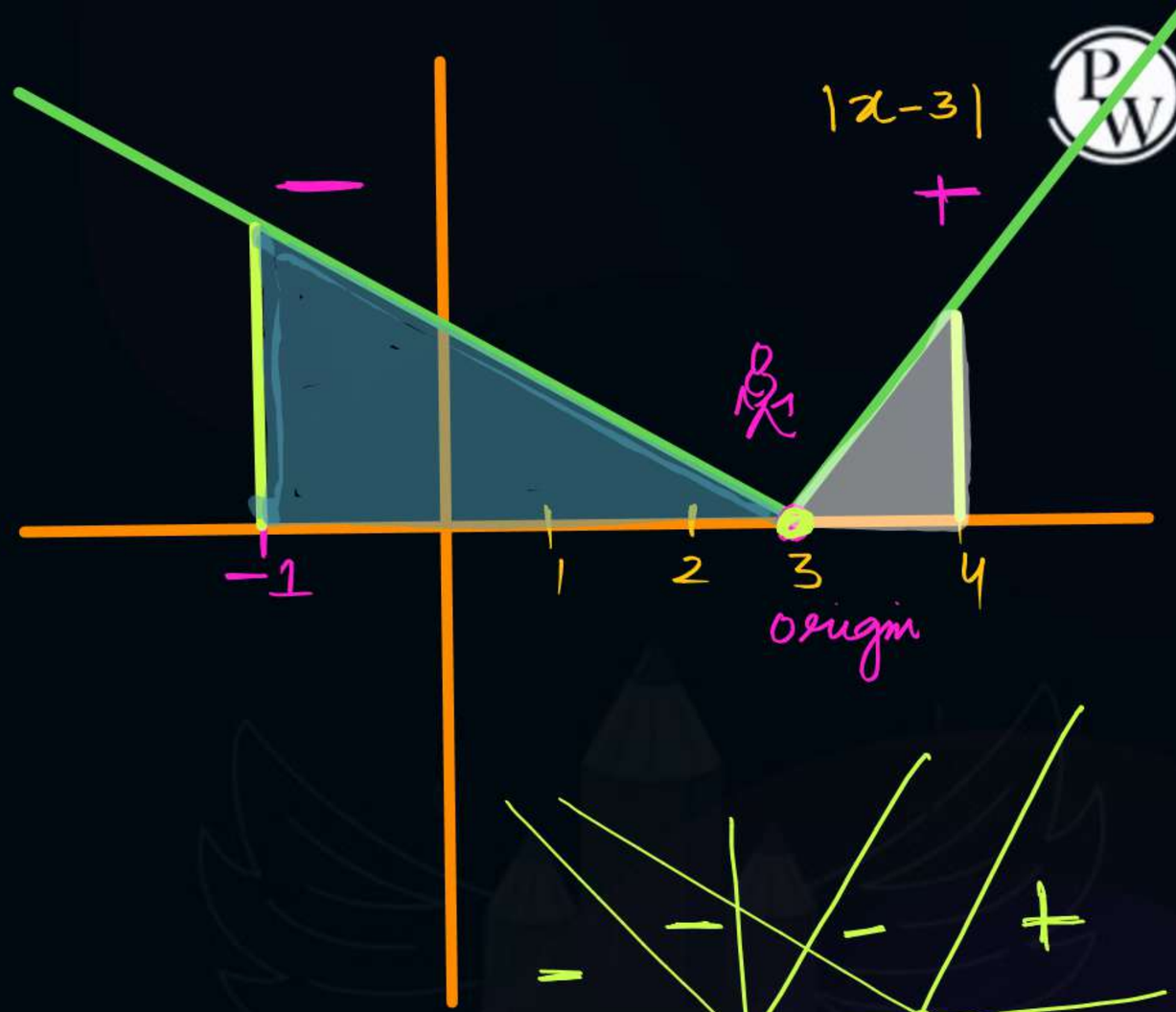
1) $$I = \int_{-1}^{4} |x-3|\,dx$$

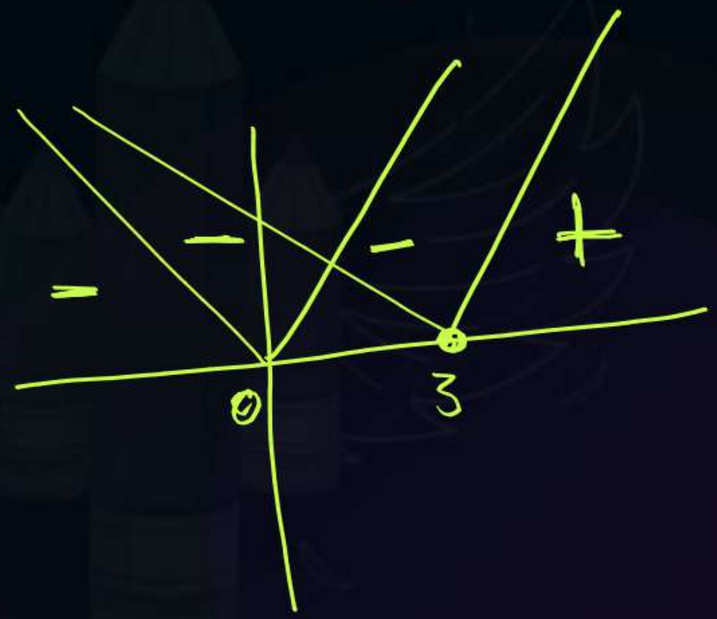
$$= \int_{-1}^{3} -(x-3)\,dx + \int_{3}^{4} +(x-3)\,dx$$

$$= \text{Ans}$$

2) $$I = \int_{-1}^{5} |x-1|\,dx$$

3) $$I = \int_{1}^{4} |x-1|\,dx$$

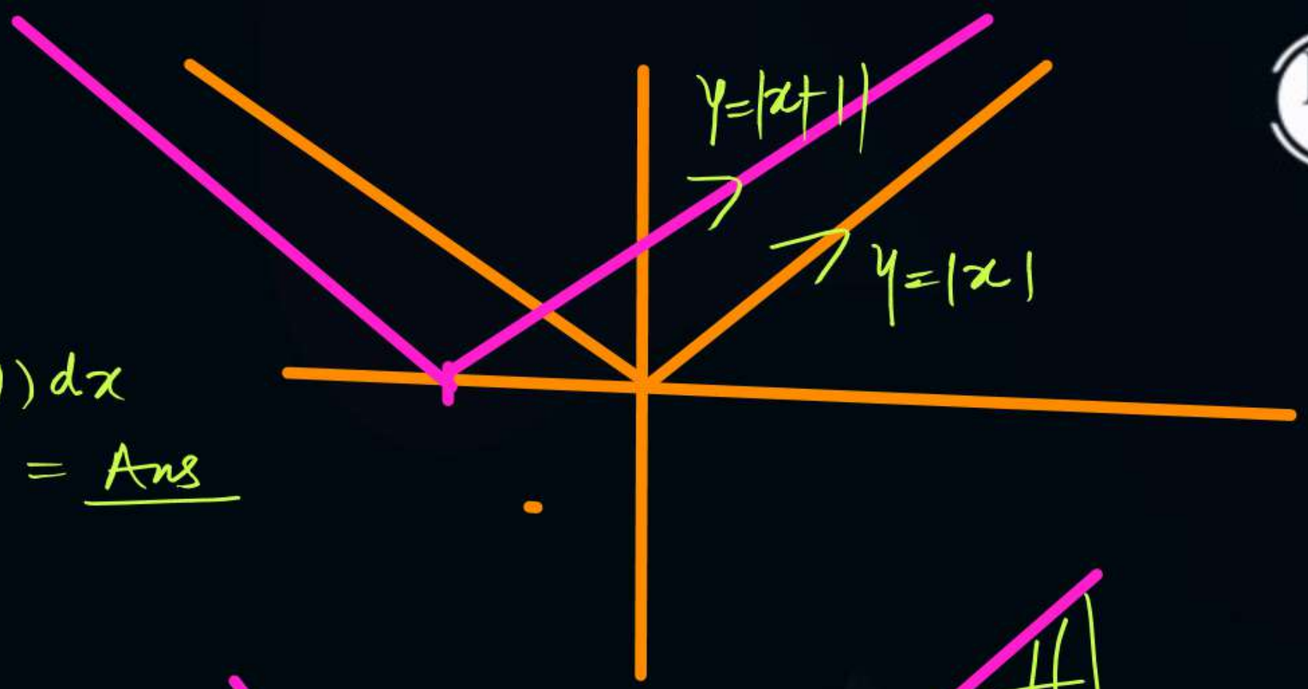




$$I = \int_{-3}^{5} |x+1|\,dx$$

$$= \int_{-3}^{-1} -(x+1)\,dx + \int_{-1}^{5} (x+1)\,dx$$

$= \underline{\text{Ans}}$





$$I = \int_0^5 [x]\,dx \qquad [1\cdot(\,)] = 1$$

$[x]$ → defined only Piecewise Function

$[x]$ defined only Integer

$$= \int_0^1 [x]\,dx + \int_1^2 [x]\,dx + \int_2^3 [x]\,dx + \int_3^4 [x]\,dx + \int_4^5 [x]\,dx$$

$$= \int_0^1 0\,dx + \int_1^2 1\,dx + \int_2^3 2\,dx + \int_3^4 3\,dx + \int_4^5 4\,dx$$

$$= 1 + 2 + 3 + 4$$

$$= 10 \quad \text{Ans.}$$

$[0] = 0$

$[0.999999] = 0$

$[1.3333] = 1$

$$I = \int_1^6 ([x]+2)\,dx = (25)$$

$$= \int_1^2 ([x]+2)\,dx + \int_2^3 ([x]+2)\,dx + \int_3^4 [x]+2\,dx + \int_4^5 ([x]+2)\,dx + \int_5^6 ([x]+2)\,dx$$

$[x]$: 1 on $[1,2)$, 2 on $[2,3)$, 3 on $[3,4)$, 4 on $[4,5)$, 5 on $[5,6)$

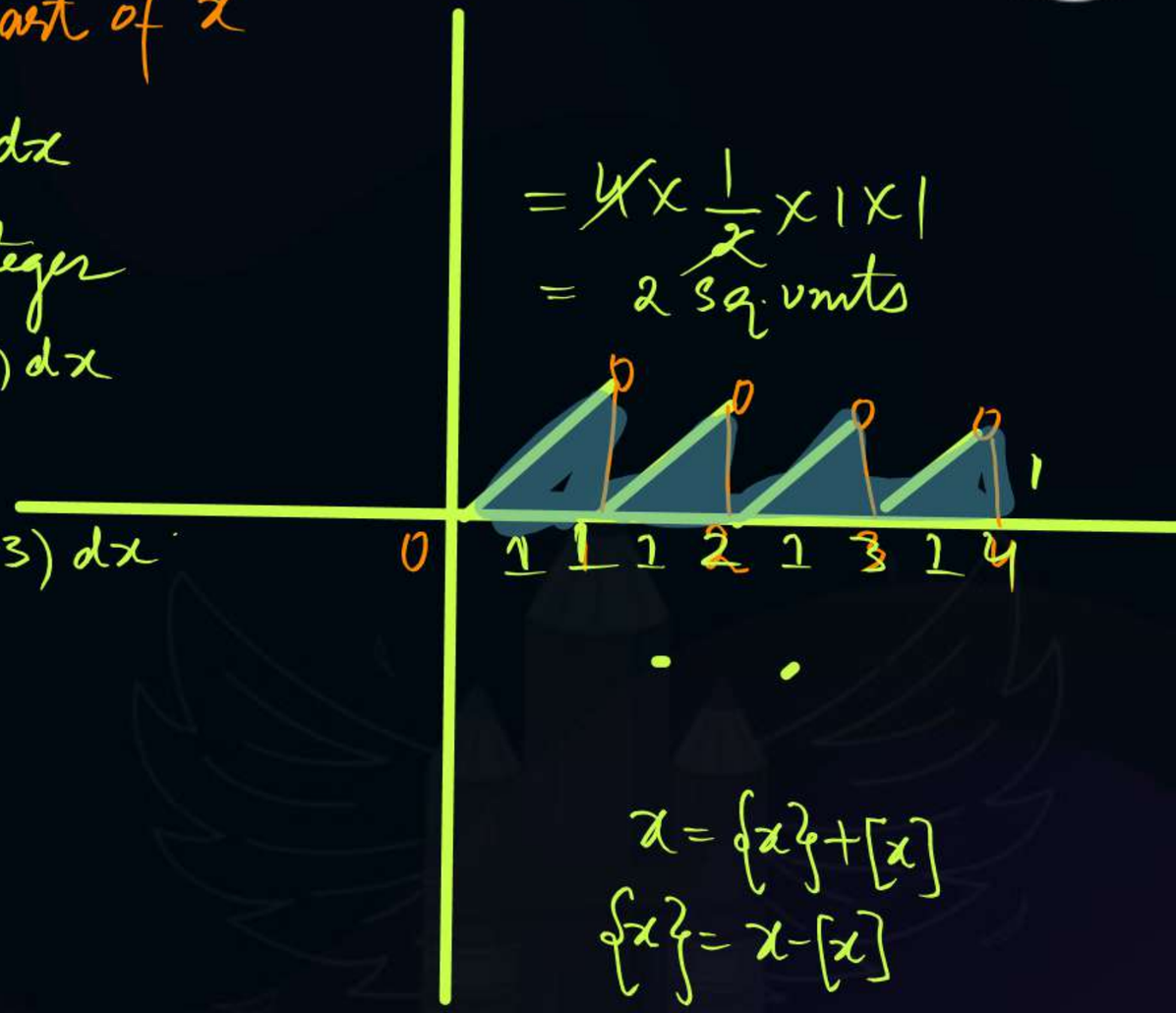
CASE 03 $f(x) = \{x\}$ = fractional Part of x

$$\int_0^4 \{x\}\,dx = \int_0^4 (x - [x])\,dx$$

($[x]$ ↙ Integer)

$$\Rightarrow \int_0^1 (x-0)\,dx + \int_1^2 (x-1)\,dx + \int_2^3 (x-2)\,dx + \int_3^4 (x-3)\,dx$$

$$= \underline{\underline{2}}$$



$= 4 \times \frac{1}{2} \times 1 \times 1$

$= 2$ sq. units

$x = \{x\} + [x]$

$\{x\} = x - [x]$

$$I = \int_0^4 (\{x\})^{[x]}\, dx$$

$$= \int_0^4 (x - [x])^{[x]}\, dx$$

$$= \int_0^1 (x-0)^0\, dx + \int_1^2 (x-1)^1\, dx + \int_2^3 (x-2)^2\, dx + \int_3^4 (x-3)^3\, dx$$

$$= \underline{\text{Ans.}}$$



2 mins Summary

Topic One → Parts (Int)

Topic Two → Definite Integrals

Topic Three

Topic Four

Topic Five

THANK - YOU

